HANIKA ANGLICAN INTEGRATED POLYTECHNIC (HAIP) · Data Mining and Data Warehousing · FA1 Group Work

Data Quality Report

Before / after metrics, cleansing results and error handling

Project: University Library Analytics Data Warehouse

Group: **Group X** (replace with your group number)

Members: Member 1 (Data Architect) · Member 2 (ETL Specialist) · Member 3 (Analytics & Security Lead)

Database: SQLite 3.53.1 · ETL: Python 3 + pandas · Dashboards: Power BI Desktop

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1. Summary

The ETL (`04_ETL_Files/etl_script.py`, batch 1) read **9,390** raw rows from the three sources, found and logged **527** data-quality problems and loaded **9,065** clean rows into the warehouse. **325** rows were rejected (mainly duplicates); every rejected row is still in the staging tables and in `etl_error_log`, so nothing is lost and every decision can be checked.

9,390	9,065	325	527	12 / 12
rows read	rows loaded	rows rejected	problems logged	validation checks passed

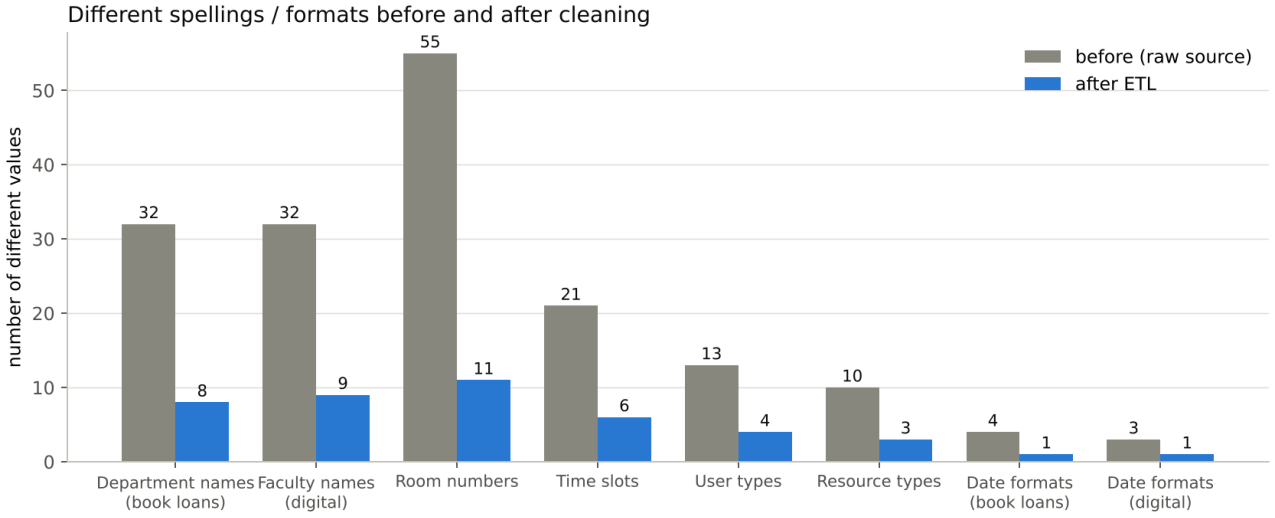


Figure 1 - every inconsistent attribute is reduced to one standard set of values

2. Before / after metrics per source

2.1 Book Lending System (book_transactions.csv)

Metric	Before	After
Rows	2,278	2,205
Exact duplicates (same TransactionID twice)	44	0
Logical duplicates (same student + book + day)	22	0
Different department spellings	32	8 departments
Date formats (incl. invalid text)	4	1 (ISO 8601)
ISBN formats (with / without dash)	2	1
Book category spellings	12	4
Missing department	79	0 (filled from student master)
Missing return date (not returned)	331	319 kept, flagged `is_returned = 0`

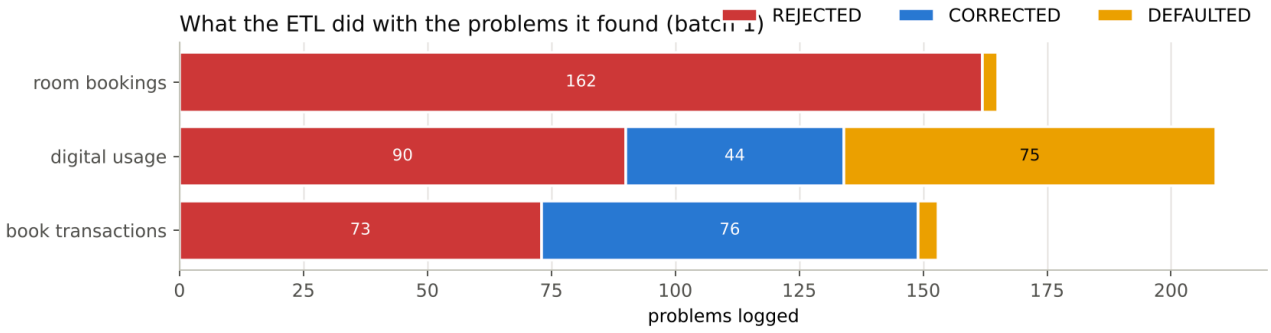
2.2 Digital Resources System (digital_usage.xlsx)

Metric	Before	After
Rows (3 sheets)	4,113	4,023
Exact duplicates	60	0
Date formats (one per sheet)	3	1
Faculty spellings	32	9 (incl. Unknown)
Resource type spellings (E-book / e-Book ...)	10	3
User type spellings	13	4
DownloadCount stored as text	77	0
Missing Duration_Minutes (legacy)	968	937 kept as NULL + flag
Missing faculty	91	0 (→ Unknown department)

2.3 Study Room Booking System (room_bookings.csv)

Metric	Before	After
Rows	2,999	2,837
Exact duplicates	112	0
Near duplicates (same room + day + slot)	43	0
Room number spellings	55	11 rooms
Time slot spellings	21	6 standard blocks
Date formats	2	1
DurationHours with text ("2 hrs", "1.5h")	1,471	0
Purpose spellings	10	6
Missing StudentID (walk-ins)	214	0 → 207 bookings point to the "Unknown" student

3. What was done with each problem



source	error type	action	rows
book_transactions	DUPLICATE_EXACT	REJECTED	44
book_transactions	DUPLICATE_LOGICAL	REJECTED	22
book_transactions	INVALID_DATE	REJECTED	4
book_transactions	INVALID_RANGE	REJECTED	3
book_transactions	MISSING_VALUE	CORRECTED	76
book_transactions	UNKNOWN_REFERENCE	DEFAULTED	4
digital_usage	DUPLICATE_EXACT	REJECTED	60

source	error type	action	rows
digital_usage	INVALID_NUMBER	REJECTED	18
digital_usage	MISSING_VALUE	DEFAULTED	75
digital_usage	SUBTOTAL_ROW	REJECTED	12
digital_usage	TEXT_NUMBER	CORRECTED	44
room_bookings	DUPLICATE_EXACT	REJECTED	112
room_bookings	DUPLICATE_NEAR	REJECTED	43
room_bookings	INVALID_DATE	REJECTED	3
room_bookings	INVALID_NUMBER	REJECTED	4
room_bookings	UNKNOWN_REFERENCE	DEFAULTED	3

Actions

- **REJECTED** - the row is not loaded (duplicates, impossible dates, invalid numbers, subtotal rows). It stays in staging and in the error log.
- **CORRECTED** - the value was fixed automatically (department filled from the student master, number stored as text converted).
- **DEFAULTED** - the value was replaced by the *Unknown* member (key 1) or NULL so the rest of the row can still be used.

One example of each error type (full list: `output/etl_error_log.csv`, readable sample: `error_log_sample.txt`):

row	source	column	bad value	error type	action
171	book_transactions	TransactionID	1169	DUPLICATE_EXACT	REJECTED
2258	book_transactions	TransactionID	3213	DUPLICATE_LOGICAL	REJECTED
2845	room_bookings	BookingID	7843	DUPLICATE_NEAR	REJECTED
877	book_transactions	CheckoutDate	2023-00-12	INVALID_DATE	REJECTED
530	digital_usage	DownloadCount	two	INVALID_NUMBER	REJECTED
257	book_transactions	ReturnDate	Apr 02, 2023	INVALID_RANGE	REJECTED
13	book_transactions	Department		MISSING_VALUE	CORRECTED
148	digital_usage	ResourceTitle	QUARTER 1 TOTAL = 1,131	SUBTOTAL_ROW	REJECTED
189	digital_usage	DownloadCount	1	TEXT_NUMBER	CORRECTED
560	book_transactions	BookISBN	978-1238724937	UNKNOWN_REFERENCE	DEFAULTED

4. Validation before loading

The load only starts when all 12 checks pass; otherwise nothing is changed.

validation check	result
No duplicate TransactionID in fact_book_loan	PASS
No duplicate BookingID in fact_room_booking	PASS
No room double-booked in the same slot	PASS
All checkout dates inside 2023-2025	PASS
All download dates inside 2023-2025	PASS
All booking dates inside 2023-2025	PASS
Loan durations are never negative	PASS

validation check	result
Download counts are never negative	PASS
Booking hours between 0 and 12	PASS
Every student_key exists in dim_student	PASS
Every book_key exists in dim_book	PASS
No NULL department_key in any fact	PASS

After loading, the ETL also checks that the row count of every warehouse table equals the prepared data and that `PRAGMA foreign_key_check` finds 0 problems. Query **Q19** in `business_queries.sql` proves there are **0 duplicates** in the final warehouse.

5. Handling of missing values - choices and reasons

Field	Missing	Strategy	Why
ReturnDate	319 loans	kept; `is_returned = 0`, `return_date_key = -1`, duration NULL	a missing date means "not returned yet" - a real business state, not an error
StudentID (rooms)	207 bookings	"Unknown" student (key 1), `is_walk_in = 1`	the booking happened; it must count in room usage
Duration_Minutes	937 rows	NULL + `duration_is_missing = 1`	we do not invent reading times; averages simply ignore NULL; the flag shows how complete the data is
Department	79 rows	looked up in the student master	the correct value exists in another system
Faculty (digital)	91 rows	Unknown department	the digital system has no student to look up

6. Error handling procedure

1. **Every run** writes one row to `etl_batch_log` (status RUNNING → SUCCESS / FAILED, rows extracted / loaded / rejected).
2. **Value-level problems** never stop the run: they are logged in `etl_error_log` (source, line number, column, bad value, error type, action, message).
3. **Batch-level problems** (a validation check fails, an SQL error, a row-count mismatch) stop the batch. Because the load is one transaction, the warehouse keeps yesterday's correct data. The reason is written to `output/etl_failure.log`.
4. **Daily review** (librarian, 5 minutes): open `error_log_sample.txt` or run `SELECT * FROM etl_error_log WHERE batch_id = (SELECT MAX(batch_id) FROM etl_batch_log)`.
5. **Fixing a recurring problem**: a new spelling → insert one row in `map_department` / `map_time_slot`; a wrong source record → correct it in the source system; then run the ETL again (it is safe to re-run: no duplicates are created).
6. **Escalation**: if a batch fails two nights in a row, the ETL specialist checks `etl_failure.log`; the dashboards keep showing the last good data meanwhile.